Assignment 2: X-ray Powder Diffraction - instrument simulation, crystallite size, and the signals view

Course: Nanotechnology & Quantum Physics | **Topic:** XRD instrument simulation, domain size, Fourier description of broadening

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Built with: Prime Agent (the AI assistant used in this session); every line of the simulator is in Appendix A.

Item	What it is
Simulator	Cu K-alpha tube, divergence and Soller slits, flat specimen on a theta/2theta goniometer, receiving slit, optional monochromator, detector
Phases	Si, Al, Cu, alpha-Fe, NaCl, MgO, ZnO, TiO2 rutile, TiO2 anatase
Physics	Bragg's law, structure factor with Waasmaier-Kirfel form factors [6], multiplicities, Lorentz-polarisation, Debye-Waller, K-alpha doublet, Caglioti resolution, Scherrer broadening, Poisson counts
Repository	https://github.com/OBODA0/quantum-physics

1. The design, and screenshots of it working (25%)

The simulator models a Bragg-Brentano powder diffractometer (Figure 1). A copper tube supplies the K-alpha doublet at 1.540598 and 1.544426 angstrom; a divergence slit and Soller slits define the beam; the flat specimen sits on a theta/2theta goniometer; a receiving slit and an optional monochromator sit in front of the detector. The measured variable is 2theta and the specimen sits at theta, so a reflection appears where 2 d sin(theta) = lambda.

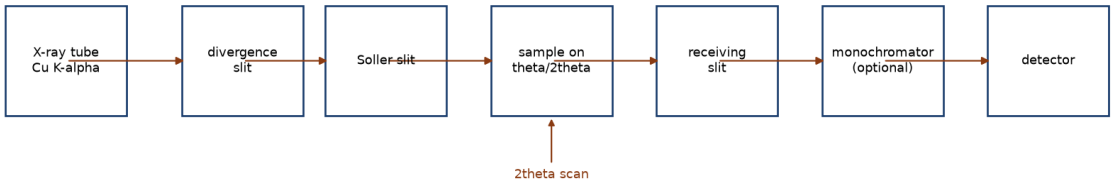


Figure 1: the chain the simulator models. Each box is a module of the code in Appendix A.

For every reflection the simulator computes a relative intensity |F|^2 m LP, where m is the powder multiplicity and LP the Lorentz-polarisation factor, then draws the line with a pseudo-Voigt profile whose width follows the Caglioti form. Counts are Poisson draws around the expected mean, so the output has the grain of a real scan. Because the absolute prefactor of a Bragg-Brentano scan is constant across reflections, intensities are relative, as on a real instrument.

1.1 Varying crystal structure

Figure 2 shows the simulator run on nine phases under one set of instrument settings. The differences are the crystallography, not the settings: an FCC metal has few strong lines and no 100; a BCC metal loses 100 and 111; silicon, with its diamond basis, loses 222 as well; rock salt has a weak 111 because the sodium and chlorine form factors nearly cancel; wurtzite splits what cubic symmetry would have degenerate; and the two titania polymorphs give different patterns from the same formula.

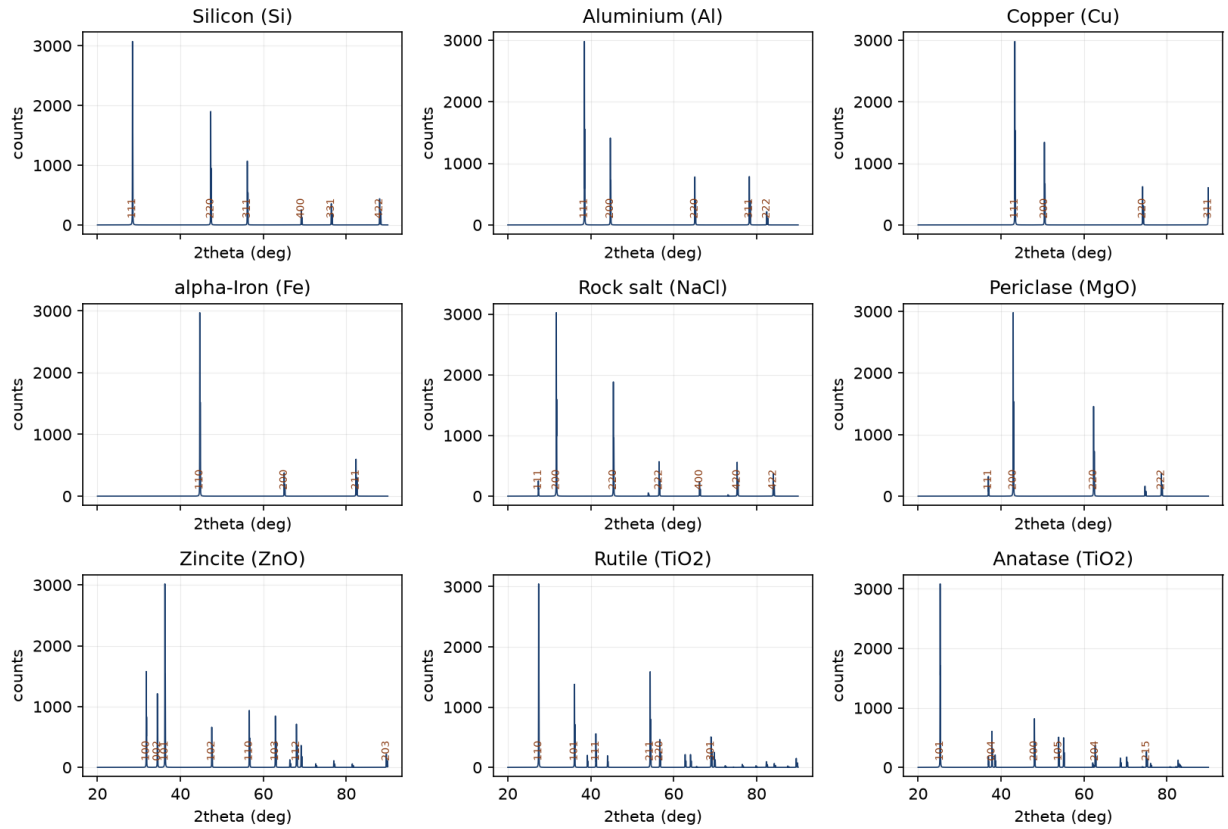


Figure 2: simulated patterns of nine phases, 20-90 deg, Cu K-alpha, 20 000 counts at the strongest line, no counting noise. The labels are the reflections the model predicts.

Key	Phase	Space group	Cell (A)	Strongest line
si	Silicon (Si)	Fd-3m (227)	a = 5.4311	111 at 28.44 deg
al	Aluminium (Al)	Fm-3m (225)	a = 4.0495	111 at 38.47 deg
cu	Copper (Cu)	Fm-3m (225)	a = 3.6149	111 at 43.32 deg
fe	alpha-Iron (Fe)	Im-3m (229)	a = 2.8665	110 at 44.67 deg
nacl	Rock salt (NaCl)	Fm-3m (225)	a = 5.6402	200 at 31.70 deg
mgo	Periclase (MgO)	Fm-3m (225)	a = 4.2112	200 at 42.92 deg
zno	Zincite (ZnO)	P63mc (186)	a = 3.2494, c = 5.2038	101 at 36.26 deg
tio2-rutile	Rutile (TiO2)	P42/mnm (136)	a = 4.5937, c = 2.9587	110 at 27.44 deg
tio2-anatase	Anatase (TiO2)	I41/amd (141)	a = 3.7845, c = 9.5143	101 at 25.31 deg

Table 1: the nine phases. The last column is the line the model predicts; cell constants and their provenance are in data/SOURCES.md, and every row is checked against its published strongest reflection in the test suite.

1.2 Varying the size of the crystalline unit

Figure 3 is the same phase, the same reflection and the same instrument, with only the crystallite size changed. The peak stays at 28.441 deg; its width grows and its height falls as the size shrinks. The right panel normalises each curve to its own maximum, which isolates the width from the height.

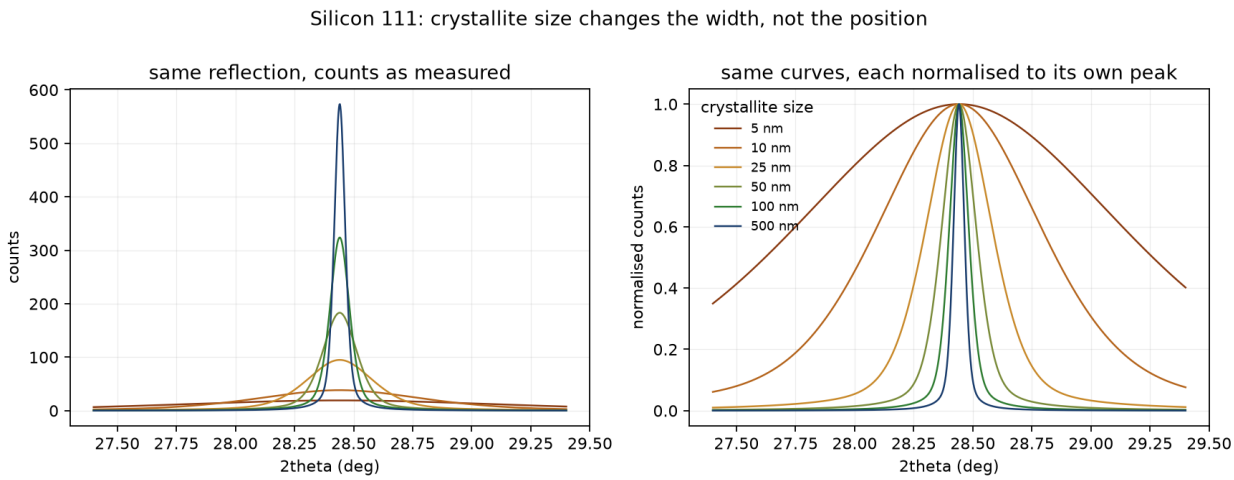


Figure 3: silicon 111 at 5, 10, 25, 50, 100 and 500 nm. Left: counts as measured. Right: each curve normalised to its own peak. Cu K-alpha1 only, so the doublet does not confuse the width.

2. What happens to the data when the domain size changes (25%)

Table 2 is the measurement behind this item: silicon, Cu K-alpha1, a 0.002 deg step, 100 000 counts at the strongest line and no counting noise, so that everything in the table is the size and nothing else.

D (nm)	FWHM 111 (deg)	FWHM 220 (deg)	FWHM 400 (deg)	height 111 (counts)	area 111 (counts deg)	Scherrer 111 (nm)
5	1.6334	1.7279	1.8931	96	148	5
10	0.8204	0.8680	0.9601	192	146	10
20	0.4130	0.4368	0.4852	382	160	20
50	0.1721	0.1816	0.2018	916	162	50
100	0.0973	0.1020	0.1133	1620	162	100
200	0.0666	0.0691	0.0765	2368	162	200
500	0.0550	0.0565	0.0624	2869	162	499

Table 2: size study. The Scherrer column subtracts the instrument width in quadrature and applies D = K lambda / (beta cos theta) with K = 0.9 [1], [3]. The area window is four FWHM wide, so at the smallest sizes it overlaps the neighbouring reflections and the area reads low.

2.1 Four things change, one thing does not

Position does not move. 111 stays at 28.441 deg for every size in Table 2: Bragg's law fixes the angle from the cell, and the cell does not change with the crystallite size.

Width grows as 1/D. FWHM at 111 goes from 0.0550 deg at 500 nm to 1.633 deg at 5 nm, a factor of 30 for a factor of 100 in size. The relation is linear in 1/D while the sample dominates: the left panel of Figure 4 is a straight line of slope -1 on log-log axes over the whole range.

Height falls, area stays. Peak height at 111 drops from 2869 counts at 500 nm to 96 counts at 5 nm, close to the 30x width factor, while the integrated area stays between 148 and 162 counts deg across the whole range. The diffracted power of that reflection is set by the structure factor and the beam, not by how the crystal is cut up; the power is redistributed into a wider peak.

The instrument sets a floor. At 500 nm the sample width (0.016 deg) is smaller than the instrument width (0.052 deg), and the measured width is the instrument's. Below roughly 100 nm the sample dominates; above it, the size can no longer be read from the width without a careful instrumental correction.

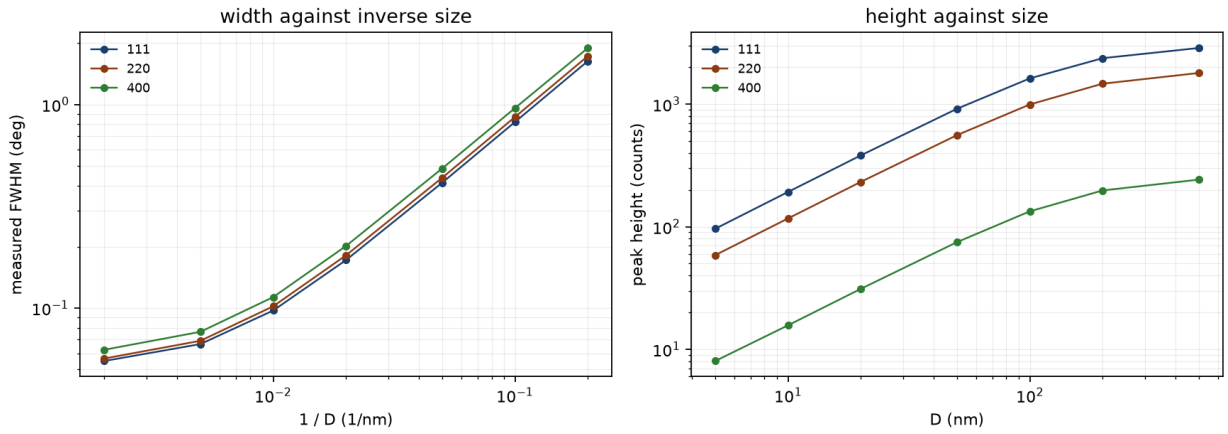


Figure 4: left, measured FWHM against 1/D on log-log axes, one line per reflection; right, peak height against D. Both curves bend over where the instrument width takes control.

2.2 The effect depends on the reflection

For one crystallite, high-angle reflections are broader than low-angle ones: at 20 nm the 111 line is 0.413 deg and the 400 line is 0.485 deg (Table 2), and Figure 5 shows the same trend over 20-120 deg. The reason is the 1/cos(theta) in the Scherrer relation, and section 3 derives it as a change of variable between reciprocal space and the goniometer angle.

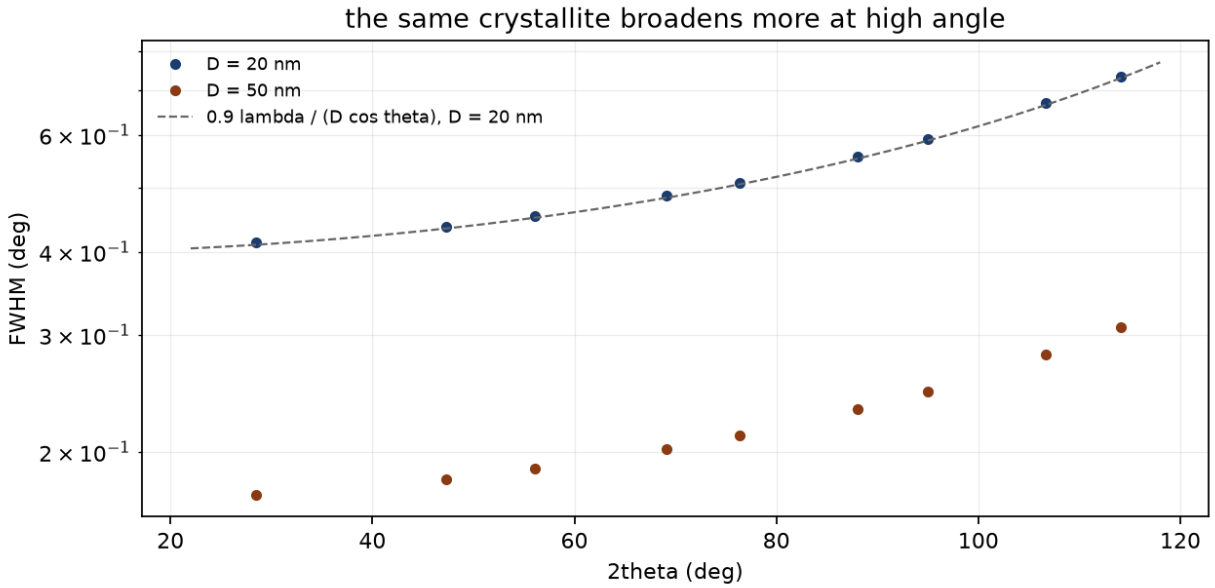


Figure 5: FWHM against 2theta for 20 nm and 50 nm crystallites, with the Scherrer curve K lambda / (D cos theta) drawn for 20 nm.

One more consequence is practical: as the peaks widen they start to overlap. In Figure 3 the 5 nm 111 peak is wider than the 220-111 separation at low angle, so neighbouring lines merge into one envelope and a peak-fitting routine that assumes separate maxima begins to fail. That is the same statement as a resolution limit in a measurement system.

3. Why it happens, in signals-and-systems terms (25%)

3.1 The pattern is a Fourier transform

The amplitude scattered by a specimen is the Fourier transform of its electron density, A(q) proportional to the integral of rho(r) exp(-i q.r) dr, with q = 4 pi sin(theta) / lambda [2, ch. 2]. An infinite crystal is a lattice convolved with a unit cell, so its transform is a reciprocal lattice multiplied by the structure factor: delta functions at the Bragg points, which is the ideal line in the top panel of Figure 6.

A real crystal is that lattice multiplied by a window w(r), equal to one inside the crystallite and zero outside. Multiplication in real space is convolution in reciprocal space [4, ch. 3], [5, ch. 2]: the amplitude is the ideal pattern convolved with W, the transform of the window, and the measured intensity carries |W|^2. For a crystal of size D the window is a rectangle and |W|^2 is a sinc-squared with its first zeros at q = 2 pi / D. The finite size of the crystal is therefore not a correction to the physics; it is a filter applied to the ideal spectrum, and its bandwidth is set by 1/D.

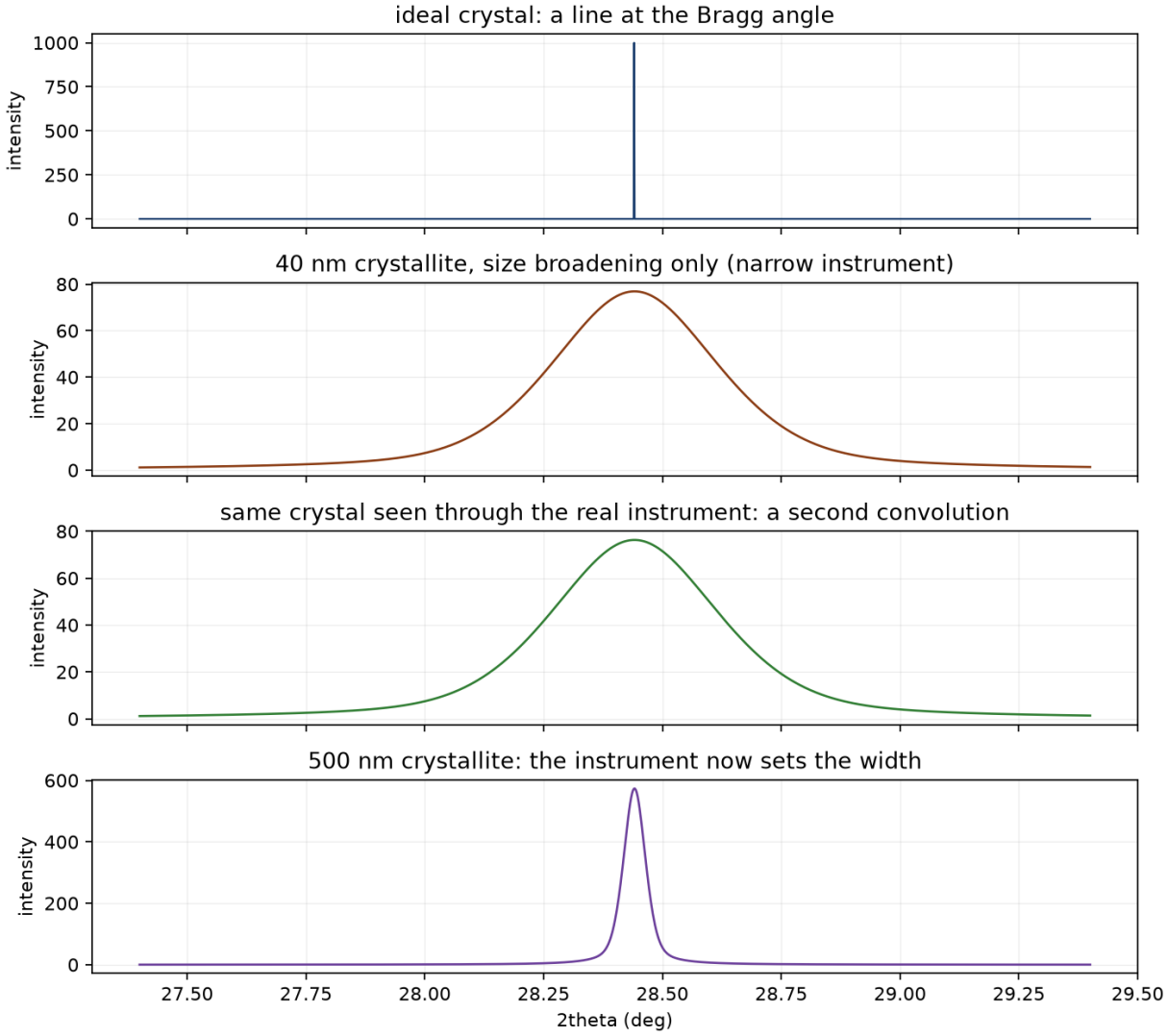


Figure 6: the same reflection at four stages. Top: an ideal, infinite crystal. Second: a 40 nm crystallite, which convolves the line with the transform of its own finite window. Third: the same crystal seen through the instrument, a second convolution. Bottom: a 500 nm crystallite, where the instrument response is again the wider of the two and therefore sets the measured width.

3.2 Two LTI stages, and what follows from that

The chain is two linear, shift-invariant stages in series. The input is the Bragg impulse train; the crystal has impulse response h_crystal = W; the instrument has impulse response h_instrument. The output is the convolution of all three. Convolution commutes and associates, so the order of the two stages does not matter and the observed line is the ideal line blurred by both responses. Two results of linear-systems theory then explain the whole of item 2 without any extra physics.

Bandwidth. A long impulse response has a narrow frequency response. A crystallite of N planes is a window of length N d, and the width of its transform scales as 1/N. Halving the crystallite doubles the width in q, which is the straight line of slope minus one in Figure 4. The reciprocal relation is the same statement as the time-bandwidth product in signals and systems: a short observation cannot resolve fine frequency structure.

Area. A normalised impulse response has unit area, and convolution preserves the integral. The area under a reflection is therefore fixed by the structure factor and the beam, while its height is area divided by width. Table 2 shows exactly that: the area at 111 stays within about ten per cent across a hundredfold change in size, while the height follows the width. In the language of the same theory this is Parseval's theorem: the total power in the line is unchanged, only its distribution in q is.

The instrument is the second filter. Its impulse response is set by the slits, the source size and the detector, and it is wider than the crystal response for any crystallite above about 100 nm. That is why the curves in Figure 4 flatten at the right: once the crystal filter is narrower than the instrument filter, the measured width stops following the size. In system terms the cascade is limited by its widest stage, just as a measurement chain is limited by its slowest element.

3.3 Why the broadening depends on the angle

The broadening happens in q, but the scan is recorded in 2theta. The two are related by q = 4 pi sin(theta) / lambda, so dq / d(2theta) = (2 pi / lambda) cos(theta). A fixed width in q therefore appears in the scan as a width

Delta(2theta) = Delta(q) lambda / (2 pi cos theta) = K lambda / (D cos theta),

which is the Scherrer relation with K = 0.9 [1]. The 1/cos(theta) is a Jacobian: it is the change of variable between the frequency axis of the spectrum and the goniometer angle on which it is displayed. At high angle a degree of 2theta spans fewer reciprocal-space units, so the same spectral width looks wider on the chart. The same variable change is why the K-alpha doublet separates more at high angle.

The scan step is the sampling rate of that displayed spectrum. Table 2 measures a 0.055 deg width at 500 nm; at a 0.02 deg step that is under three points per FWHM, and the measured width becomes sensitive to where the samples land. The 0.002 deg step used for Table 2 is chosen for that reason: to sample the narrowest line of the study rather than to chase resolution the instrument does not have.

3.4 Does the prediction hold?

The Scherrer curve drawn in Figure 5 is not fitted: it is K lambda / (D cos theta) with the same K and lambda the simulator uses. At 20 nm it predicts 0.410 deg for 111 and the measured width is 0.413 deg; at 400 it predicts 0.482 and measures 0.485. At 5 nm the prediction is 1.64 deg against 1.63 measured. The agreement is a check that the model's size convolution is the transform argument above and not an empirical fit.

4. Verification, and what the model does not do

The simulator is covered by a test suite; the numbers quoted here are produced by the same code that draws the figures, and the report is gated on them. Peak positions are checked against the NIST SRM 640f certificate, every phase against its published strongest line, and the whole report rebuilds from a fresh clone byte-identically. The suite, the gates and the provenance of every constant are documented in the repository.

Not modelled: Rietveld refinement, anomalous dispersion, preferred orientation, K-beta residual, detector energy discrimination, thin-film absorption and absolute intensities. Every site shares one Debye-Waller factor. The Cu K-alpha wavelengths are the conventional values; no primary table was read. None of these affects the size argument, which depends only on the width and the convolution.

References

[1] A. L. Patterson, *The Scherrer formula for X-ray particle size determination*, Physical Review **56**, 978-982 (1939). doi:10.1103/PhysRev.56.978

[2] B. E. Warren, *X-ray Diffraction*, Addison-Wesley (1969); Dover reprint (1990), ISBN 0-486-66317-5. Chapters 2 and 3: scattering by a finite lattice, the interference function.

[3] B. D. Cullity and S. R. Stock, *Elements of X-ray Diffraction*, 3rd edition, Prentice Hall (2001), ISBN 0-201-61091-4. Chapters 4 and 5: instrument broadening and crystallite-size measurement.

[4] A. V. Oppenheim and A. S. Willsky, *Signals and Systems*, 2nd edition, Prentice Hall (1997), ISBN 0-13-814757-4. Convolution, the convolution theorem, and the frequency response of a linear system.

[5] R. N. Bracewell, *The Fourier Transform and Its Applications*, 3rd edition, McGraw-Hill (2000), ISBN 0-07-303938-1. Transform pairs for the rectangle and the finite chain, and the scaling rule.

[6] D. Waasmaier and A. Kirfel, *New analytical scattering-factor functions for free atoms and ions*, Acta Crystallographica **A51**, 416-431 (1995). doi:10.1107/S0108767394013292

Every entry was checked on 2026-10-22: the DOI resolves through Crossref, the ISBNs resolve through OpenLibrary, and each work is cited for the claim it supports.

Appendix A: the code

Every module the simulator ships, in alphabetical order. The same files are in the repository under assignments/02-xrd-diffractometer/xrdsim/.

xrdsim/__init__.py

```
"""XRD diffractometer simulator (assignment 2, quantum physics section)."""
__version__ = "0.1.0"
```

xrdsim/__main__.py

```
from .cli import main

raise SystemExit(main())
```

xrdsim/analysis.py

```
"""Reading a simulated or measured pattern: peak finding, widths and indexing.

The functions here work on arrays (2theta, counts) and make no assumption about where
they came from, so the same code is used on simulator output and on a measured scan.
"""

from __future__ import annotations

import math
from dataclasses import dataclass

import numpy as np

from .profile import scherrer_size


@dataclass(frozen=True)
class Peak:
    two_theta: float
    height: float
    fwhm: float
    area: float


def find_peaks(two_theta: np.ndarray, counts: np.ndarray,
               min_height_fraction: float = 0.02) -> list[Peak]:
    """Local maxima above a fraction of the tallest point, with width and area."""
    x = np.asarray(two_theta, dtype=float)
    y = np.asarray(counts, dtype=float)
    if x.size != y.size or x.size < 3:
        raise ValueError("need matching arrays of at least three points")
    threshold = min_height_fraction * y.max()
    peaks: list[Peak] = []
    for i in range(1, len(y) - 1):
        if not (y[i] >= y[i - 1] and y[i] > y[i + 1] and y[i] >= threshold):
            continue
        centre, height = _refine(x, y, i)
        half = (height + _local_floor(x, y, i)) / 2.0
        fwhm = _width_at(x, y, i, half)
        area = _area(x, y, i, 4.0 * fwhm if fwhm > 0 else x[1] - x[0])
        peaks.append(Peak(centre, height, fwhm, area))
    return peaks


def _local_floor(x: np.ndarray, y: np.ndarray, i: int, reach_deg: float = 10.0) -> float:
    """Baseline near a peak: the 10th percentile within `reach_deg`, so a neighbouring
    reflection cannot drag the half-height down and the result does not depend on the step."""
    step = x[1] - x[0] if len(x) > 1 else 1.0
    reach = max(1, int(reach_deg / step))
    left = max(0, i - reach)
    right = min(len(y), i + reach)
    return float(np.percentile(y[left:right], 10))


def _width_at(x: np.ndarray, y: np.ndarray, i: int, half: float) -> float:
```

```
    left = i
    while left > 0 and y[left] > half:
        left -= 1
    right = i
    while right < len(y) - 1 and y[right] > half:
        right += 1
    if left == i or right == i:
        return 0.0
    x_left = np.interp(half, [y[left], y[left + 1]], [x[left], x[left + 1]])
    x_right = np.interp(half, [y[right], y[right - 1]], [x[right], x[right - 1]])
    return float(x_right - x_left)


def _area(x: np.ndarray, y: np.ndarray, i: int, half_window: float) -> float:
    window = np.abs(x - x[i]) <= half_window
    return float(np.trapezoid(y[window] - np.median(y[window]), x[window]))


def pure_width(observed: float, instrumental: float) -> float:
    """Sample-only FWHM by Gaussian quadrature subtraction, guarded at zero."""
    value = observed * observed - instrumental * instrumental
    return math.sqrt(value) if value > 0 else 0.0


def scherrer_from_peak(peak: Peak, instrumental_fwhm: float, wavelength: float,
                       k: float = 0.9) -> float:
    """Crystallite size in angstrom from one peak, corrected for instrumental width."""
    beta = pure_width(peak.fwhm, instrumental_fwhm)
    if beta <= 0:
        raise ValueError("the peak is not broader than the instrument")
    return scherrer_size(beta, peak.two_theta, wavelength, k)


def index_cubic(two_thetas: list[float],
                max_n_squared: int = 30, tolerance: float = 0.02) -> list[tuple[tuple[int, int, int], float]]:
    """Assign (hkl) to a cubic pattern by the sin^2(theta) ratios.

    The cell returned is the smallest one that fits, so a body-centred pattern comes back
    as a primitive cell with half the conventional edge; `lattice_hint` names that.

    Returns the assignments and the implied n^2 = h^2 + k^2 + l^2. The first reflection
    is taken as the lowest-angle one; every other angle must fall on an integer multiple
    of its sin^2(theta) to within `tolerance` (relative).
    """
    if not two_thetas:
        raise ValueError("no peaks to index")
    two_thetas = sorted(two_thetas)
    s2 = [math.sin(math.radians(t / 2.0)) ** 2 for t in two_thetas]
    base = s2[0]
    for n2_first in range(1, max_n_squared + 1):
        ratios = [value / base * n2_first for value in s2]
        if all(abs(r - round(r)) <= tolerance * max(round(r), 1) for r in ratios):
            assignments = []
            for angle, ratio in zip(two_thetas, ratios):
                n2 = int(round(ratio))
                assignments.append((_hkl_from_n2(n2), angle))
            return assignments
    raise ValueError("the pattern does not index as cubic")


def lattice_hint(n_squared: list[int]) -> str:
    """Describe the lattice from the observed n^2 sequence.

    All-even n^2 means a body-centred lattice described on half the cell; the FCC sequence
    starts 3, 4, 8, 11, 12, 16, 19, 20, 24. Anything else is primitive.
    """
    values = sorted({int(value) for value in n_squared})
    fcc = {3, 4, 8, 11, 12, 16, 19, 20, 24, 27, 32, 35, 36, 40, 43, 44, 48}
    if values and all(value % 2 == 0 for value in values):
        return "body-centred, conventional cell twice the refined edge"
    if set(values) <= fcc:
        return "face-centred (the 3, 4, 8, 11 sequence)"
    if 1 in values:
        return "primitive cell; a body-centred sample would have twice this edge"
    return "primitive lattice"


def _hkl_from_n2(n2: int) -> tuple[int, int, int]:
    for h in range(0, 8):
        for k in range(0, 8):
            l_sq = n2 - h * h - k * k
            if l_sq < 0:
                continue
            l = round(math.isqrt(l_sq))
```

```
            if l * l == l_sq:
                return tuple(sorted((h, k, l), reverse=True))
    raise ValueError(f"no (hkl) has n^2 = {n2}")


def _refine(x: np.ndarray, y: np.ndarray, i: int) -> tuple[float, float]:
    """Parabolic refinement of a sampled maximum: position and height between bins."""
    y0, y1, y2 = y[i - 1], y[i], y[i + 1]
    denom = y0 - 2.0 * y1 + y2
    if denom == 0:
        return float(x[i]), float(y[i])
    delta = 0.5 * (y0 - y2) / denom
    if abs(delta) > 0.5:
        return float(x[i]), float(y[i])
    return float(x[i] + delta * (x[i + 1] - x[i])), float(y1 - 0.25 * (y0 - y2) * delta)


def williamson_hall(peaks: list[Peak], instrumental_fwhm_for,
                    wavelength: float, k: float = 0.9) -> tuple[float, float]:
    """Crystallite size and microstrain from the Williamson-Hall plot.

    beta cos(theta) = K lambda / D + 4 epsilon sin(theta), fitted by least squares over the
    peaks given. `instrumental_fwhm_for(two_theta)` returns the instrument width at that
    angle. Returns (size in angstrom, strain) and raises when fewer than two peaks are
    usable.
    """
    xs, ys = [], []
    for peak in peaks:
        beta = pure_width(peak.fwhm, instrumental_fwhm_for(peak.two_theta))
        if beta <= 0 or peak.two_theta <= 0:
            continue
        theta = math.radians(peak.two_theta / 2.0)
        xs.append(4.0 * math.sin(theta))
        ys.append(math.radians(beta) * math.cos(theta))
    if len(xs) < 2:
        raise ValueError("Williamson-Hall needs at least two peaks broader than the instrument")
    slope, intercept = np.polyfit(xs, ys, 1)
    if intercept <= 0:
        raise ValueError("the fit has no positive intercept, so no size can be reported")
    return k * wavelength / intercept, float(slope)


def identify_phase(two_thetas: list[float], phase, instrument,
                   tolerance: float = 0.15,
                   min_intensity_fraction: float = 0.15) -> float:
    """Fraction of a phase's strong reflections that the observed peaks contain.

    A score of 1.0 means every predicted line above `min_intensity_fraction` of the
    phase's strongest has an observed peak within `tolerance` degrees.
    """
    from .diffractometer import reflections
    rows = reflections(phase, instrument)
    if not rows:
        return 0.0
    strongest = max(row.intensity for row in rows)
    predicted = [row.two_theta for row in rows
                 if row.intensity > min_intensity_fraction * strongest]
    if not predicted or not two_thetas:
        return 0.0
    matched = sum(1 for angle in predicted
                  if min(abs(angle - seen) for seen in two_thetas) <= tolerance)
    return matched / len(predicted)


def identify(two_thetas: list[float], phases: dict, instrument,
             tolerance: float = 0.15) -> list[tuple[str, float]]:
    """Rank the phases of a database by how well they explain the observed peak positions."""
    scored = [(key, identify_phase(two_thetas, phase, instrument, tolerance))
              for key, phase in phases.items()]
    return sorted(scored, key=lambda row: (-row[1], row[0]))
```

xrdsim/atoms.py

```
"""Atomic scattering factors and the Debye-Waller term.

`f0` is interpolated from data/scattering_factors.json, sampled once from the
Waasmaier-Kirfel analytic form factors (Acta Cryst. A51, 416-431, 1995).
`s = sin(theta) / lambda` in 1/angstrom.
"""

from __future__ import annotations

import json
```

```
import math
import pathlib

DATA_PATH = pathlib.Path(__file__).resolve().parent.parent / "data" / "scattering_factors.json"

_table: dict | None = None


def _load() -> dict:
    global _table
    if _table is None:
        payload = json.loads(DATA_PATH.read_text(encoding="utf-8"))
        _table = {
            "s_max": payload["grid_s_inv_A"][-1],
            "step": payload["grid_s_inv_A"][1] - payload["grid_s_inv_A"][0],
            "elements": payload["f0"],
        }
    return _table


def f0(element: str, s: float) -> float:
    """Elastic form factor f0 in electrons. `s = sin(theta)/lambda` in 1/angstrom."""
    table = _load()
    values = table["elements"].get(element)
    if values is None:
        raise KeyError(f"no scattering factor for {element!r}; rebuild data/scattering_factors.json")
    if s < 0:
        raise ValueError("s must be non-negative")
    position = s / table["step"]
    last = len(values) - 1
    if position >= last:
        return float(values[last])
    i = int(position)
    frac = position - i
    return float(values[i] * (1.0 - frac) + values[i + 1] * frac)


def debye_waller(b_iso: float, s: float) -> float:
    """Temperature factor exp(-B s^2), with B = 8 pi^2 <u^2> in square angstrom."""
    if b_iso < 0:
        raise ValueError("B must be non-negative")
    return math.exp(-b_iso * s * s)
```

xrdsim/cli.py

```
"""Command line interface: `python -m xrdsim`."""

from __future__ import annotations

import argparse
import pathlib

from . import sources
from .analysis import (find_peaks, identify, index_cubic, lattice_hint,
                       scherrer_from_peak, williamson_hall)
from .diffractometer import Instrument, reflections, scan
from .profile import caglioti_fwhm
from .samples import PHASES, get_phase


def _instrument(args) -> Instrument:
    return Instrument(
        ka2=not args.no_ka2,
        monochromator_two_theta=args.monochromator,
        reference_counts=args.counts,
        background_rate=args.background,
        dwell_s=args.dwell,
        rng_seed=None if args.noise == 0 else args.noise,
        sample_displacement_mm=args.displacement,
    )


def main(argv: list[str] | None = None) -> int:
    parser = argparse.ArgumentParser(prog="xrdsim", description=__doc__)
    parser.add_argument("--phase", help="phase key (see --list-phases)")
    parser.add_argument("--range", nargs=2, type=float, metavar=("MIN", "MAX"),
                        default=[20.0, 90.0], help="2theta scan range in degrees")
    parser.add_argument("--step", type=float, default=0.02, help="2theta step in degrees")
    parser.add_argument("--dwell", type=float, default=1.0, help="counting time per step, s")
    parser.add_argument("--counts", type=float, default=20_000.0,
                        help="integrated counts at the strongest reflection")
    parser.add_argument("--background", type=float, default=0.5, help="background, counts/s")
    parser.add_argument("--displacement", type=float, default=0.0, metavar="MM",
```

```
                      help="sample surface displaced from the goniometer axis")
    parser.add_argument("--size", type=float, default=None,
                      help="crystallite size in angstrom (adds size broadening)")
    parser.add_argument("--no-ka2", action="store_true", help="drop the K-alpha2 line")
    parser.add_argument("--monochromator", type=float, default=None,
                      help="monochromator 2theta in degrees")
    parser.add_argument("--noise", type=int, default=0,
                      help="Poisson seed; 0 disables counting noise")
    parser.add_argument("--out", type=pathlib.Path, help="write the scan to this image")
    parser.add_argument("--table", action="store_true", help="print the reflection table")
    parser.add_argument("--index", action="store_true",
                      help="index the observed peaks as a cubic pattern")
    parser.add_argument("--scherrer", action="store_true",
                      help="report Scherrer sizes for the observed peaks")
    parser.add_argument("--identify", action="store_true",
                      help="rank the phase database against the simulated pattern")
    parser.add_argument("--list-phases", action="store_true", help="list the phase database")
    args = parser.parse_args(argv)

    if args.list_phases:
        for key, phase in sorted(PHASES.items()):
            print(f"{key:14s} {phase.formula:8s} {phase.space_group:18s} {phase.source}")
        return 0
    if not args.phase:
        parser.error("--phase is required unless --list-phases is given")

    try:
        return _run(args)
    except (KeyError, ValueError) as error:
        parser.error(str(error).strip('"'))


def _run(args) -> int:
        phase = get_phase(args.phase)
        instrument = _instrument(args)
        result = scan(phase, args.range[0], args.range[1], args.step, instrument,
                      size_a=args.size)

        if args.table:
            strongest = max(row.intensity for row in result.reflections)
            print(f"{'hkl':>10s} {'d (A)':>8s} {'2theta':>8s} {'m':>3s} {'I/I0':>7s}")
            for row in result.reflections:
                if row.intensity < 1e-4 * strongest:
                    continue
                label = "".join(str(v) for v in row.hkl)
                print(f"{label:>10s} {row.d:8.4f} {row.two_theta:8.3f} {row.multiplicity:3d} "
                      f"{100 * row.intensity / strongest:7.2f}")
            return 0

        peaks = find_peaks(result.two_theta, result.counts)
        print(f"phase      : {phase.name} ({phase.formula}), {phase.space_group}")
        print(f"wavelength : {instrument.wavelength:.6f} A"
              + (" + K-alpha2" if instrument.ka2 else ""))
        print(f"range      : {args.range[0]} - {args.range[1]} deg, step {args.step}")
        print(f"peaks      : {len(peaks)}")
        for peak in peaks[:12]:
            line = f"  {peak.two_theta:8.3f} deg  height {peak.height:9.0f}  FWHM {peak.fwhm:.4f}"
            if args.size and not args.scherrer:
                try:
                    size = scherrer_from_peak(peak, caglioti_fwhm(peak.two_theta, *instrument.caglioti),
                                              instrument.wavelength)
                    line += f"  Scherrer {size / 10.0:.1f} nm"
                except ValueError:
                    pass
            print(line)

        if args.index:
            try:
                rows = index_cubic([peak.two_theta for peak in peaks])
            except ValueError as error:
                print(f"indexing failed: {error}")
            else:
                print("indexing   : hkl            2theta")
                for hkl, angle in rows:
                    print(f"             {str(hkl):14s} {angle:7.3f}")
                n_squared = [hkl[0] ** 2 + hkl[1] ** 2 + hkl[2] ** 2 for hkl, _ in rows]
                print(f"lattice    : {lattice_hint(n_squared)}")

        if args.scherrer:
            print("Scherrer   : peak           size (nm)")
            for peak in peaks:
                try:
                    size = scherrer_from_peak(
                        peak, caglioti_fwhm(peak.two_theta, *instrument.caglioti),
```

```
                    instrument.wavelength)
            except ValueError:
                continue
            print(f"             {peak.two_theta:7.3f}         {size / 10.0:6.1f}")
        try:
            wh_size, wh_strain = williamson_hall(
                peaks, lambda angle: caglioti_fwhm(angle, *instrument.caglioti),
                instrument.wavelength)
        except ValueError as error:
            print(f"Williamson-Hall: {error}")
        else:
            print(f"Williamson-Hall: D = {wh_size / 10.0:.1f} nm, strain = {wh_strain:.5f}"
                  + ("" if args.no_ka2 else " (use --no-ka2: the shoulders bias the fit)"))

    if args.identify:
        ranking = identify([peak.two_theta for peak in peaks], PHASES, instrument)
        print("identification: phase          score")
        for key, score in ranking[:5]:
            print(f"               {key:14s} {score:5.2f}")

    if args.out:
        from .plot import save_scan
        args.out.parent.mkdir(parents=True, exist_ok=True)
        save_scan(result, args.out,
                  title=f"{phase.name}, {instrument.wavelength:.4f} A")
        print(f"wrote {args.out}")
    return 0
```

xrdsim/detector.py

```
"""Detector: counting statistics and dead time.

A powder scan is a counting experiment. The model returns expected counts per step; the
detector turns them into observed counts with Poisson noise.
"""

from __future__ import annotations

import math


def expected_counts(intensity: float, dwell_s: float) -> float:
    """Mean counts at a step: intensity (counts per second) times dwell time."""
    if dwell_s <= 0:
        raise ValueError("dwell time must be positive")
    return max(intensity, 0.0) * dwell_s


def observe(mean_counts: float, rng) -> float:
    """One Poisson draw for a step. `rng` is a numpy Generator."""
    return float(rng.poisson(max(mean_counts, 0.0)))


def count_noise(mean_counts: float) -> float:
    """Standard deviation of a Poisson count (sqrt(N)); zero for an empty step."""
    return math.sqrt(max(mean_counts, 0.0))


def apply_dead_time(true_rate: float, tau_s: float) -> float:
    """Observed rate for a non-paralysable detector: n = m / (1 + m tau)."""
    if true_rate < 0 or tau_s < 0:
        raise ValueError("rates and dead time must be non-negative")
    return true_rate / (1.0 + true_rate * tau_s)
```

xrdsim/diffractometer.py

```
"""The diffractometer: reflections, peak profiles and the scan itself.

A scan is built in counts per step. Reflection intensities are relative (I/I0), as in
Assignment 1: the model reproduces patterns and intensity ratios, not absolute
radiometry.
"""

from __future__ import annotations

import math
from dataclasses import dataclass, field

import numpy as np

from . import sources
from .detector import apply_dead_time, expected_counts
from .lattice import d_spacing
```

```
from .profile import (add_in_quadrature, caglioti_fwhm, pseudo_voigt_array,
                      size_fwhm)
from .samples import Phase
from .structure import lorentz_polarisation, multiplicity, structure_factor


@dataclass(frozen=True)
class Instrument:
    wavelength: float = sources.CU_KA1
    ka2: bool = True
    monochromator_two_theta: float | None = None
    caglioti: tuple[float, float, float] = (0.004, -0.002, 0.003)
    eta: float = 0.5
    reference_counts: float = 20_000.0     # counts at the strongest reflection
    background_rate: float = 0.5           # counts per second, flat
    dwell_s: float = 1.0
    dead_time_s: float = 0.0
    rng_seed: int | None = None      # None: no counting noise; an int seeds the Poisson draws
    max_two_theta: float = 150.0      # no goniometer reaches 180 deg, and LP diverges there
    sample_displacement_mm: float = 0.0  # specimen surface off the goniometer axis
    goniometer_radius_mm: float = 240.0


@dataclass(frozen=True)
class Reflection:
    hkl: tuple[int, int, int]
    d: float
    two_theta: float
    multiplicity: int
    f_squared: float
    intensity: float


@dataclass
class Scan:
    two_theta: np.ndarray
    counts: np.ndarray
    reflections: list[Reflection] = field(default_factory=list)
    phase: Phase | None = None
    instrument: Instrument | None = None



def reflections(phase: Phase, instrument: Instrument, hmax: int = 8) -> list[Reflection]:
    """Every reflection the wavelength reaches, with its relative intensity."""
    rows: list[Reflection] = []
    for h in range(0, hmax + 1):
        for k in range(0, hmax + 1):
            for l in range(0, hmax + 1):
                if h == 0 and k == 0 and l == 0:
                    continue
                if not _representative(h, k, l, phase.system):
                    continue
                try:
                    d = d_spacing(h, k, l, **phase.cell_kwargs())
                except ValueError:
                    continue
                if 2.0 * d < instrument.wavelength:
                    continue
                two_theta = 2.0 * math.degrees(
                    math.asin(instrument.wavelength / (2.0 * d)))
                if two_theta > instrument.max_two_theta:
                    continue
                f = structure_factor(phase, h, k, l, instrument.wavelength)
                m = multiplicity(h, k, l, phase.system)
                lp = lorentz_polarisation(two_theta, instrument.monochromator_two_theta)
                rows.append(Reflection((h, k, l), d, two_theta, m, abs(f) ** 2,
                                       abs(f) ** 2 * m * lp))
    rows.sort(key=lambda r: r.two_theta)
    return rows


def scan(phase: Phase, two_theta_min: float, two_theta_max: float, step: float,
         instrument: Instrument | None = None, size_a: float | None = None,
         hmax: int = 8) -> Scan:
    """Simulate one 2theta scan, with Poisson noise and optional crystallite size."""
    if instrument is None:
        instrument = Instrument()
    if step <= 0 or two_theta_max <= two_theta_min:
        raise ValueError("the scan range and step must be positive")
    grid = np.arange(two_theta_min, two_theta_max + step / 2.0, step)
    rows = reflections(phase, instrument, hmax)
    if not rows:
        raise ValueError("no reachable reflections in this range")
```

```
    strongest = max(row.intensity for row in rows)
    if strongest <= 0:
        raise ValueError("every reflection has zero intensity")

    lines = [(instrument.wavelength, 1.0)]
    if instrument.ka2:
        # the doublet is scaled from the instrument's own K-alpha1 so that another anode
        # keeps its own wavelengths instead of silently getting copper's
        ka2 = instrument.wavelength * sources.CU_KA2 / sources.CU_KA1
        lines = [(instrument.wavelength, 1.0), (ka2, sources.KA2_TO_KA1)]

    mean = np.full_like(grid, expected_counts(instrument.background_rate, instrument.dwell_s),
                        dtype=float)
    for row in rows:
        area_total = instrument.reference_counts * row.intensity / strongest
        for wavelength, weight in lines:
            d = d_spacing(*row.hkl, **phase.cell_kwargs())
            centre = 2.0 * math.degrees(math.asin(min(1.0, wavelength / (2.0 * d))))
            centre += displacement_shift(centre, instrument.sample_displacement_mm,
                                         instrument.goniometer_radius_mm)
            width = caglioti_fwhm(centre, *instrument.caglioti)
            if size_a is not None:
                width = add_in_quadrature(width, size_fwhm(size_a, centre, wavelength))
            if not (two_theta_min - 5.0 * width <= centre <= two_theta_max + 5.0 * width):
                continue
            profile = pseudo_voigt_array(grid, centre, width, instrument.eta)
            mean += area_total * weight * profile * step

    if instrument.dead_time_s:
        rate = mean / instrument.dwell_s
        mean = np.array([apply_dead_time(r, instrument.dead_time_s) * instrument.dwell_s
                         for r in rate])

    if instrument.rng_seed is None:
        counts = mean
    else:
        rng = np.random.default_rng(instrument.rng_seed)
        counts = rng.poisson(np.clip(mean, 0.0, None)).astype(float)

    return Scan(grid, counts, rows, phase, instrument)


def _representative(h: int, k: int, l: int, system: str) -> bool:
    """True for one member of each symmetry family.

    Enumerating h, k, l >= 0 visits every permutation of a reflection; the multiplicity
    already counts the whole orbit, so only the canonical member may be used.
    """
    if system == "cubic":
        return h >= k >= l
    if system == "tetragonal":
        return h >= k
    if system == "hexagonal":
        return h >= k
    return True


def displacement_shift(two_theta_deg: float, displacement_mm: float,
                       radius_mm: float) -> float:
    """2theta shift in degrees for a flat specimen displaced from the goniometer axis.

    Delta(2theta) = -2 s cos(theta) / R, in radians.
    """
    if displacement_mm == 0.0:
        return 0.0
    if radius_mm <= 0:
        raise ValueError('the goniometer radius must be positive')
    theta = math.radians(two_theta_deg / 2.0)
    return math.degrees(-2.0 * displacement_mm * math.cos(theta) / radius_mm)
```

xrdsim/geometry.py

```
"""Bragg-Brentano geometry: sample illumination and absorption.

Angles in degrees, lengths in mm, mass attenuation in cm^2/g. The flat-plate,
infinite-thickness case is the one a laboratory powder scan approximates.
"""

from __future__ import annotations

import math
def bragg_angle(two_theta_deg: float) -> float:
    """theta for a given 2theta."""
```

```
    return two_theta_deg / 2.0


def footprint_length(two_theta_deg: float, divergence_slit_mm: float,
                     sample_length_mm: float = 20.0) -> float:
    """Length of sample illuminated by the beam, in mm.

    The beam leaves the divergence slit with a fixed width, so the footprint on the
    flat plate grows as 1/sin(theta) and is clipped by the sample holder.
    """
    theta = math.radians(bragg_angle(two_theta_deg))
    if theta <= 0:
        raise ValueError("2theta must be positive")
    return min(sample_length_mm, divergence_slit_mm / math.sin(theta))


def linear_attenuation(density_g_cm3: float, mass_attenuation_cm2_g: float) -> float:
    """mu in 1/cm."""
    if density_g_cm3 <= 0 or mass_attenuation_cm2_g <= 0:
        raise ValueError("density and mass attenuation must be positive")
    return density_g_cm3 * mass_attenuation_cm2_g


def is_infinitely_thick(mu_per_cm: float, thickness_cm: float, threshold: float = 0.01) -> bool:
    """True when exp(-2 mu t / sin(theta)) is negligible; here the worst case theta -> 90 deg."""
    return math.exp(-2.0 * mu_per_cm * thickness_cm) < threshold


def flat_plate_absorption(mu_per_cm: float) -> float:
    """Absorption factor for a thick flat plate: A = 1 / (2 mu), independent of theta."""
    if mu_per_cm <= 0:
        raise ValueError("mu must be positive")
    return 1.0 / (2.0 * mu_per_cm)


def penetration_depth(mu_per_cm: float, two_theta_deg: float, fraction: float = 0.99) -> float:
    """Depth in cm that carries `fraction` of the diffracted intensity."""
    if not 0.0 < fraction < 1.0:
        raise ValueError("fraction must lie in (0, 1)")
    theta = math.radians(bragg_angle(two_theta_deg))
    return -math.log(1.0 - fraction) * math.sin(theta) / (2.0 * mu_per_cm)
```

xrdsim/lattice.py

```
"""Lattice geometry: d-spacings, Bragg angles and lattice-parameter refinement.

Lengths are in angstrom, angles in degrees. Only the settings a powder scan needs are
supported here: cubic, tetragonal, hexagonal and orthorhombic.
"""

from __future__ import annotations

import math


def d_spacing(h: int, k: int, l: int, *, a: float, b: float | None = None,
              c: float | None = None, hexagonal: bool = False) -> float:
    """Interplanar spacing in angstrom.

    `hexagonal=True` selects the hexagonal setting (a = b, gamma = 120 degrees).
    Otherwise `b` and `c` both None is cubic, `c` given with `b` None is tetragonal, and
    both `b` and `c` given is orthorhombic. A lone `b` is rejected rather than guessed.
    """
    if a <= 0:
        raise ValueError("lattice parameter must be positive")
    if hexagonal:
        if c is None:
            raise ValueError("the hexagonal setting needs c")
        inv_sq = 4.0 / 3.0 * (h * h + h * k + k * k) / a**2 + l * l / c**2
    elif b is None and c is None:
        inv_sq = (h * h + k * k + l * l) / a**2
    elif b is None:
        inv_sq = (h * h + k * k) / a**2 + l * l / c**2
    elif c is None:
        raise ValueError("an orthorhombic cell needs both b and c; for a tetragonal cell pass c and leave b No
    ne")
    else:
        inv_sq = h * h / a**2 + k * k / b**2 + l * l / c**2
    if inv_sq == 0:
        raise ValueError("(0,0,0) has no d-spacing")
    return 1.0 / math.sqrt(inv_sq)
def two_theta(d: float, wavelength: float) -> float:
    """Bragg angle in degrees for a d-spacing, or ValueError if the reflection is unreachable."""
```

```
    if d <= 0:
        raise ValueError("d-spacing must be positive")
    s = wavelength / (2.0 * d)
    if s > 1.0:
        raise ValueError(
            f"d = {d:.4f} A is unreachable at lambda = {wavelength:.4f} A (needs sin(theta) > 1)")
    return math.degrees(2.0 * math.asin(s))


def d_from_two_theta(two_theta_deg: float, wavelength: float) -> float:
    """d-spacing in angstrom from a measured 2theta, by Bragg's law."""
    theta = math.radians(two_theta_deg / 2.0)
    s = math.sin(theta)
    if s == 0:
        raise ValueError("2theta = 0 carries no information")
    return wavelength / (2.0 * s)


def refine_cubic_a(peaks: list[tuple[float, int, int, int]], wavelength: float) -> float:
    """Least-squares cubic lattice parameter from indexed peaks.

    `peaks` is a list of (two_theta_deg, h, k, l). The fit is an unweighted least squares
    through the origin, sin^2(theta) = (lambda^2 / 4a^2) * (h^2 + k^2 + l^2): no per-peak
    uncertainties, and no zero-offset term, so a displaced specimen biases the result.
    """
    xs, ys = [], []
    for two_theta_deg, h, k, l in peaks:
        n_sq = h * h + k * k + l * l
        if n_sq == 0:
            raise ValueError("(0,0,0) cannot be indexed")
        xs.append(n_sq)
        ys.append(math.sin(math.radians(two_theta_deg / 2.0)) ** 2)
    denom = sum(x * x for x in xs)
    if denom == 0:
        raise ValueError("no peaks given")
    slope = sum(x * y for x, y in zip(xs, ys)) / denom
    if slope <= 0:
        raise ValueError("degenerate peak list")
    return wavelength / (2.0 * math.sqrt(slope))


def peak_positions(hkls, wavelength: float, **cell) -> list[tuple[tuple[int, int, int], float, float]]:
    """(hkl, d, 2theta) for every reflection the wavelength can reach, sorted by 2theta."""
    out = []
    for h, k, l in hkls:
        d = d_spacing(h, k, l, **cell)
        try:
            out.append(((h, k, l), d, two_theta(d, wavelength)))
        except ValueError:
            continue
    return sorted(out, key=lambda row: row[2])
```

xrdsim/plot.py

```
"""Matplotlib rendering of a scan.

Assignment 1 already fixed the house style: a scan is drawn as counts against 2theta,
with the reflections the model predicts marked, and nothing decorated that the data do
not carry.
"""

from __future__ import annotations

import matplotlib

matplotlib.use("Agg")

import matplotlib.pyplot as plt

from .diffractometer import Scan


def plot_scan(scan: Scan, ax=None, title: str | None = None, annotate: bool = True):
    """Draw one scan; returns the axes."""
    if ax is None:
        _, ax = plt.subplots(figsize=(9, 4.2))
    ax.plot(scan.two_theta, scan.counts, lw=0.8, color="#1a3d6d")
    if annotate and scan.reflections:
        strongest = max(row.intensity for row in scan.reflections)
        placed: list[float] = []
        for row in sorted(scan.reflections, key=lambda r: -r.intensity):
            if row.intensity < 0.08 * strongest:
                continue
```

```
            if any(abs(row.two_theta - angle) < 1.2 for angle in placed):
                continue
            placed.append(row.two_theta)
            ax.annotate("".join(str(v) for v in row.hkl), (row.two_theta, 0),
                        textcoords="offset points", xytext=(0, 4),
                        fontsize=6.5, rotation=90, ha="center", va="bottom",
                        color="#8a3b12")
    ax.set_xlabel("2theta (deg)")
    ax.set_ylabel("counts")
    if title:
        ax.set_title(title)
    ax.grid(alpha=0.25, lw=0.5)
    return ax


def save_scan(scan: Scan, path, title: str | None = None, dpi: int = 160) -> None:
    """Write one scan to an image file."""
    fig, ax = plt.subplots(figsize=(9, 4.2))
    plot_scan(scan, ax=ax, title=title or (scan.phase.name if scan.phase else None))
    fig.tight_layout()
    fig.savefig(path, dpi=dpi)
    plt.close(fig)
```

xrdsim/profile.py

```
"""Peak profiles and the instrument resolution function.

The profile is normalised to unit area, so a peak's height follows from its integrated
intensity and its width. Angles are in degrees, widths are FWHM in degrees.
"""

from __future__ import annotations

import math

TWO_LN2 = 2.0 * math.log(2.0)


def gaussian(x: float, center: float, fwhm: float) -> float:
    """Unit-area Gaussian."""
    sigma = fwhm / (2.0 * math.sqrt(TWO_LN2))
    z = (x - center) / sigma
    return math.exp(-0.5 * z * z) / (sigma * math.sqrt(2.0 * math.pi))


def lorentzian(x: float, center: float, fwhm: float) -> float:
    """Unit-area Lorentzian."""
    half = fwhm / 2.0
    return (half / math.pi) / ((x - center) ** 2 + half * half)


def pseudo_voigt(x: float, center: float, fwhm: float, eta: float) -> float:
    """Unit-area pseudo-Voigt: eta of Lorentzian mixed with (1 - eta) of Gaussian."""
    if not 0.0 <= eta <= 1.0:
        raise ValueError("eta must lie in [0, 1]")
    return eta * lorentzian(x, center, fwhm) + (1.0 - eta) * gaussian(x, center, fwhm)


def caglioti_fwhm(two_theta_deg: float, U: float, V: float, W: float) -> float:
    """Instrumental FWHM in degrees: U tan^2(theta) + V tan(theta) + W."""
    t = math.tan(math.radians(two_theta_deg / 2.0))
    value = U * t * t + V * t + W
    return math.sqrt(max(value, 0.0))


def scherrer_size(fwhm_deg: float, two_theta_deg: float, wavelength: float,
                  k: float = 0.9) -> float:
    """Crystallite size in angstrom from a sample-broadened FWHM (Scherrer).

    `fwhm_deg` must already be corrected for instrumental broadening.
    """
    if fwhm_deg <= 0:
        raise ValueError("FWHM must be positive")
    beta = math.radians(fwhm_deg)
    theta = math.radians(two_theta_deg / 2.0)
    return k * wavelength / (beta * math.cos(theta))


def size_fwhm(size_a: float, two_theta_deg: float, wavelength: float,
              k: float = 0.9) -> float:
    """FWHM in degrees produced by a crystallite of `size_a` angstrom (inverse Scherrer)."""
    if size_a <= 0:
        raise ValueError("size must be positive")
```

```
    theta = math.radians(two_theta_deg / 2.0)
    return math.degrees(k * wavelength / (size_a * math.cos(theta)))


def add_in_quadrature(*fwhms: float) -> float:
    """Combine Gaussian widths."""
    return math.sqrt(sum(f * f for f in fwhms))


def pseudo_voigt_array(x, center: float, fwhm: float, eta: float):
    """Unit-area pseudo-Voigt over an array; same numbers as pseudo_voigt, vectorised."""
    import numpy as np
    if not 0.0 <= eta <= 1.0:
        raise ValueError("eta must lie in [0, 1]")
    values = np.asarray(x, dtype=float)
    sigma = fwhm / (2.0 * math.sqrt(TWO_LN2))
    half = fwhm / 2.0
    z = (values - center) / sigma
    gaussian_part = np.exp(-0.5 * z * z) / (sigma * math.sqrt(2.0 * math.pi))
    lorentzian_part = (half / math.pi) / ((values - center) ** 2 + half * half)
    return eta * lorentzian_part + (1.0 - eta) * gaussian_part
```

xrdsim/samples.py

```
"""The phase database: cell, symmetry-expanded basis and the source of each number.

Coordinates are fractions of the cell edge. Every site list below is the full basis of
the conventional cell, already expanded by the space-group operations, so the structure
factor needs no symmetry engine. Lattice constants carry the record they came from.
"""

from __future__ import annotations

from dataclasses import dataclass, field


@dataclass(frozen=True)
class Site:
    element: str
    x: float
    y: float
    z: float
    occupancy: float = 1.0
    b_iso: float = 0.5          # square angstrom; a stated assumption for every phase


@dataclass(frozen=True)
class Phase:
    key: str
    name: str
    formula: str
    space_group: str
    a: float
    c: float | None = None
    b: float | None = None
    hexagonal: bool = False
    sites: tuple[Site, ...] = field(default_factory=tuple)
    source: str = ""

    @property
    def system(self) -> str:
        if self.hexagonal:
            return "hexagonal"
        if self.b is not None:
            return "orthorhombic"
        return "cubic" if self.c is None else "tetragonal"

    def cell_kwargs(self) -> dict:
        return {"a": self.a, "b": self.b, "c": self.c, "hexagonal": self.hexagonal}

    @property
    def volume(self) -> float:
        """Conventional-cell volume in cubic angstrom."""
        if self.hexagonal:
            return self.a * self.a * self.c * 0.8660254037844386
        if self.b is None and self.c is None:
            return self.a ** 3
        return self.a * (self.b or self.a) * self.c


def _fcc(element: str, b_iso: float = 0.5) -> tuple[Site, ...]:
    return tuple(Site(element, x, y, z, b_iso=b_iso) for x, y, z in
                 [(0, 0, 0), (0, .5, .5), (.5, 0, .5), (.5, .5, 0)])
```

```
def _rocksalt(cation: str, anion: str) -> tuple[Site, ...]:
    sites = list(_fcc(cation))
    sites += [Site(anion, x + .5, y, z) for x, y, z in
              [(0, 0, 0), (0, .5, .5), (.5, 0, .5), (.5, .5, 0)]]
    return tuple(sites)


PHASES: dict[str, Phase] = {
    "si": Phase(
        key="si", name="Silicon", formula="Si", space_group="Fd-3m (227)",
        a=5.431144, sites=tuple(
            [Site("Si", x, y, z) for x, y, z in
             [(0, 0, 0), (0, .5, .5), (.5, 0, .5), (.5, .5, 0),
              (.25, .25, .25), (.25, .75, .75), (.75, .25, .75), (.75, .75, .25)]]),
        source="NIST SRM 640f certificate, a = 5.431144(8) A at 22.5 C"),
    "al": Phase(
        key="al", name="Aluminium", formula="Al", space_group="Fm-3m (225)",
        a=4.0495, sites=_fcc("Al"),
        source="room-temperature conventional value 4.0495 A; the AFLOW Al1_ICSD_52611 primitive (2.856 A) con
    verts to 4.0398 A, a different entry"),
    "cu": Phase(
        key="cu", name="Copper", formula="Cu", space_group="Fm-3m (225)",
        a=3.6149, sites=_fcc("Cu"),
        source="room-temperature conventional value 3.6149 A; AFLOW Cu1_ICSD_53755 converts to 3.6146 A"),
    "fe": Phase(
        key="fe", name="alpha-Iron", formula="Fe", space_group="Im-3m (229)",
        a=2.8665, sites=(Site("Fe", 0, 0, 0), Site("Fe", .5, .5, .5)),
        source="room-temperature conventional value; the AFLOW ICSD entry reads 2.9647 A and is flagged as exp
    anded"),
    "nacl": Phase(
        key="nacl", name="Rock salt", formula="NaCl", space_group="Fm-3m (225)",
        a=5.6402, sites=_rocksalt("Na", "Cl"),
        source="room-temperature conventional value 5.6402 A; AFLOW Cl1Na1_ICSD_60280 converts to 5.665 A"),
    "mgo": Phase(
        key="mgo", name="Periclase", formula="MgO", space_group="Fm-3m (225)",
        a=4.2112, sites=_rocksalt("Mg", "O"),
        source="room-temperature conventional value 4.2112 A; AFLOW Mg1O1_ICSD_173128 converts to 4.253 A"),
    "zno": Phase(
        key="zno", name="Zincite", formula="ZnO", space_group="P63mc (186)",
        a=3.2494, c=5.2038, hexagonal=True, sites=(
            Site("Zn", 1/3, 2/3, 0.0), Site("Zn", 2/3, 1/3, 0.5),
            Site("O", 1/3, 2/3, 0.3830), Site("O", 2/3, 1/3, 0.8830)),
        source="COD 9004180, a = 3.2494, c = 5.2038 A; u(O) = 0.3830"),
    "tio2-rutile": Phase(
        key="tio2-rutile", name="Rutile", formula="TiO2", space_group="P42/mnm (136)",
        a=4.5937, c=2.9587, sites=(
            Site("Ti", 0, 0, 0), Site("Ti", .5, .5, .5),
            Site("O", .3047, .3047, 0), Site("O", .6953, .6953, 0),
            Site("O", .8047, .1953, .5), Site("O", .1953, .8047, .5)),
        source="room-temperature conventional values a = 4.5937, c = 2.9587 A; u(O) = 0.3047. COD 9004143 is a
     600 C record, which is why its cell reads 4.616/2.977 A"),
    "tio2-anatase": Phase(
        key="tio2-anatase", name="Anatase", formula="TiO2", space_group="I41/amd (141)",
        a=3.7845, c=9.5143, sites=(
            Site("Ti", 0, 0, 0), Site("Ti", 0, .5, .25), Site("Ti", .5, .5, .5), Site("Ti", .5, 0, .75),
            Site("O", 0, 0, .2079), Site("O", 0, .5, .4579), Site("O", .5, .5, .7079),
            Site("O", .5, 0, .9579), Site("O", 0, .5, .0421), Site("O", 0, 0, .7921),
            Site("O", .5, 0, .5421), Site("O", .5, .5, .2921)),
        source="room-temperature cell a = 3.7845, c = 9.5143 A; z(O) = 0.2079 with the basis expanded by gemmi
     from COD 9008214, a 300 C record"),
}


def get_phase(key: str) -> Phase:
    try:
        return PHASES[key]
    except KeyError:
        raise KeyError(f"unknown phase {key!r}; choose from {', '.join(sorted(PHASES))}") from None
```

xrdsim/sources.py

```
"""X-ray source: the copper tube and its emission lines.

Wavelengths in angstrom. The K-alpha doublet is what a powder pattern shows; K-beta is
normally removed by a filter or a monochromator and is only modelled to show why.
"""

from __future__ import annotations

from dataclasses import dataclass

CU_KA1 = 1.540598
CU_KA2 = 1.544426
```

```
CU_KB = 1.392250

# K-alpha2 / K-alpha1 intensity ratio, and K-beta / K-alpha ratio for a Cu tube.
KA2_TO_KA1 = 0.497
KB_TO_KA = 0.20


@dataclass(frozen=True)
class Line:
    wavelength: float
    weight: float


def emission_lines(ka2: bool = True, kbeta: bool = False) -> list[Line]:
    """Emission lines of a Cu tube with their relative intensities."""
    lines = [Line(CU_KA1, 1.0)]
    if ka2:
        lines.append(Line(CU_KA2, KA2_TO_KA1))
    if kbeta:
        lines.append(Line(CU_KB, KB_TO_KA * (1.0 + (KA2_TO_KA1 if ka2 else 0.0))))
    return lines


def weighted_wavelength(lines: list[Line]) -> float:
    """Intensity-weighted mean wavelength of the lines in use."""
    total = sum(line.weight for line in lines)
    if total <= 0:
        raise ValueError("no lines")
    return sum(line.wavelength * line.weight for line in lines) / total
```

xrdsim/structure.py

```
"""Structure factors, multiplicities and systematic absences.

A phase carries its cell and an explicit, symmetry-expanded basis; this module turns
that into |F(hkl)|^2 and the multiplicity the powder average needs.
"""

from __future__ import annotations

import cmath
import math

from .atoms import debye_waller, f0
from .lattice import d_spacing


def structure_factor(phase, h: int, k: int, l: int, wavelength: float) -> complex:
    """F(hkl) in electrons, including the Debye-Waller term of every site."""
    d = d_spacing(h, k, l, **phase.cell_kwargs())
    s = 1.0 / (2.0 * d)                # sin(theta)/lambda, by Bragg's law
    total = 0j
    for site in phase.sites:
        phase_angle = 2.0 * math.pi * (h * site.x + k * site.y + l * site.z)
        total += (site.occupancy * f0(site.element, s) * debye_waller(site.b_iso, s)
                  * cmath.exp(1j * phase_angle))
    return total


def multiplicity(h: int, k: int, l: int, system: str) -> int:
    """Number of symmetry-equivalent planes in the powder average."""
    h, k, l = abs(h), abs(k), abs(l)
    if system == "cubic":
        perms = 6 if len({h, k, l}) == 3 else (3 if len({h, k, l}) == 2 else 1)
        return perms * 2 ** sum(v != 0 for v in (h, k, l))
    if system == "tetragonal":
        hk = 2 if h != k else 1
        return (hk * 2 ** sum(v != 0 for v in (h, k))) * (2 if l else 1)
    if system == "hexagonal":
        if h == 0 and k == 0:
            return 2 if l else 1
        # (h, k) on a mirror plane (h == 0, k == 0 or h == k) has a 6-member in-plane orbit;
        # a general (h, k) has 12. A non-zero l doubles either.
        in_plane = 6 if (h == 0 or k == 0 or h == k) else 12
        return 2 * in_plane if l else in_plane
    raise ValueError(f"multiplicity is not defined for {system!r}")


def lorentz_polarisation(two_theta_deg: float, monochromator_two_theta: float | None = None) -> float:
    """LP factor for an unpolarised tube, optionally with a monochromator."""
    t = math.radians(two_theta_deg)
    theta = t / 2.0
    cos2t = math.cos(t) ** 2
```

```
    if monochromator_two_theta is None:
        numerator = 1.0 + cos2t
    else:
        numerator = 1.0 + cos2t * math.cos(math.radians(monochromator_two_theta)) ** 2
    return numerator / (math.sin(theta) ** 2 * math.cos(theta))


def relative_intensity(phase, h: int, k: int, l: int, wavelength: float,
                       monochromator_two_theta: float | None = None) -> float:
    """I/I0 for one reflection: |F|^2 m LP, normalised later across the pattern."""
    f = structure_factor(phase, h, k, l, wavelength)
    d = d_spacing(h, k, l, **phase.cell_kwargs())
    two_theta = math.degrees(2.0 * math.asin(wavelength / (2.0 * d)))
    m = multiplicity(h, k, l, phase.system)
    return abs(f) ** 2 * m * lorentz_polarisation(two_theta, monochromator_two_theta)
```